

ALLIANCE UNIVERSITY

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NAAC GRADE A+ ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Data Cleaning Using Power Query

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Data Cleaning Using Power Query

GitHub Link:https://github.com/PragnaChelsiya/BussinessAnalytics-Projects/tree/main/Experiment%204

1. Objective

The objective of this task is to clean and prepare a customer sales dataset using Power Query before performing further analysis. The dataset contains missing values, duplicate records, and inconsistent entries such as ERROR and UNKNOWN.

2. Dataset

The dataset used for this task is **dirty_cafe_sales.csv**.

The dataset contains the following columns:

- Transaction ID
- Item
- Quantity
- Price Per Unit
- Total Spent
- Payment Method
- Location
- Transaction Date

3. Data Cleaning Performed

3.1 Duplicate Records

Duplicate transaction records were checked using the **Transaction ID** column. Duplicate records were removed using:

Home → Remove Rows → Remove Duplicates

3.2 Missing Values

Missing values were handled according to the type of data.

For categorical columns such as **Item, Payment Method, and Location**, missing values were replaced with **Unknown**.

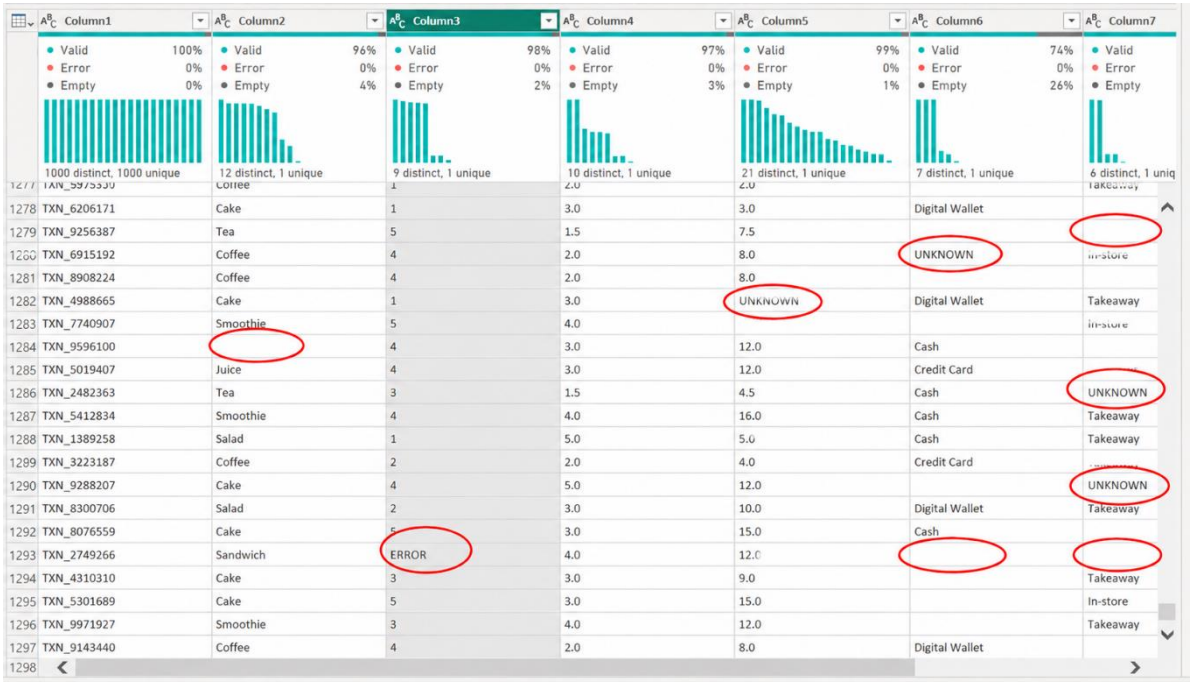
For numeric and date columns such as **Quantity, Price Per Unit, Total Spent, and Transaction Date**, unavailable values were kept as **null** instead of entering incorrect values.

3.3 Inconsistent Entries

The dataset contained inconsistent values such as:

- ERROR
- UNKNOWN
- Blank values

These entries were standardized. ERROR and UNKNOWN values in categorical columns were changed to **Unknown**, while invalid numeric and date values were handled as **null**.



	Column1	Column2	Column3	Column4	Column5	Column6	Column7
	Valid 100% Error 0% Empty 0%	Valid 96% Error 0% Empty 4%	Valid 98% Error 0% Empty 2%	Valid 97% Error 0% Empty 3%	Valid 99% Error 0% Empty 1%	Valid 74% Error 0% Empty 26%	Valid Error Empty
	1000 distinct, 1000 unique	12 distinct, 1 unique	9 distinct, 1 unique	10 distinct, 1 unique	21 distinct, 1 unique	7 distinct, 1 unique	6 distinct, 1 uniq
1278	TXN_6206171	Cake	1	3.0	3.0	Digital Wallet	
1279	TXN_9256387	Tea	5	1.5	7.5		
1280	TXN_6915192	Coffee	4	2.0	8.0	UNKNOWN	
1281	TXN_8908224	Coffee	4	2.0	8.0		
1282	TXN_4988665	Cake	1	3.0	UNKNOWN	Digital Wallet	Takeaway
1283	TXN_7740907	Smoothie	5	4.0			
1284	TXN_9596100		4	3.0	12.0	Cash	
1285	TXN_5019407	Juice	4	3.0	12.0	Credit Card	
1286	TXN_2482363	Tea	3	1.5	4.5	Cash	UNKNOWN
1287	TXN_5412834	Smoothie	4	4.0	16.0	Cash	Takeaway
1288	TXN_1389258	Salad	1	5.0	5.0	Cash	Takeaway
1289	TXN_3223187	Coffee	2	2.0	4.0	Credit Card	
1290	TXN_9288207	Cake	4	5.0	12.0		UNKNOWN
1291	TXN_8300706	Salad	2	3.0	10.0	Digital Wallet	Takeaway
1292	TXN_8076559	Cake		3.0	15.0	Cash	
1293	TXN_2749266	Sandwich	ERROR	4.0	12.0		
1294	TXN_4310310	Cake	3	3.0	9.0		Takeaway
1295	TXN_5301689	Cake	5	3.0	15.0		In-store
1296	TXN_9971927	Smoothie	3	4.0	12.0		Takeaway
1297	TXN_9143440	Coffee	4	2.0	8.0	Digital Wallet	

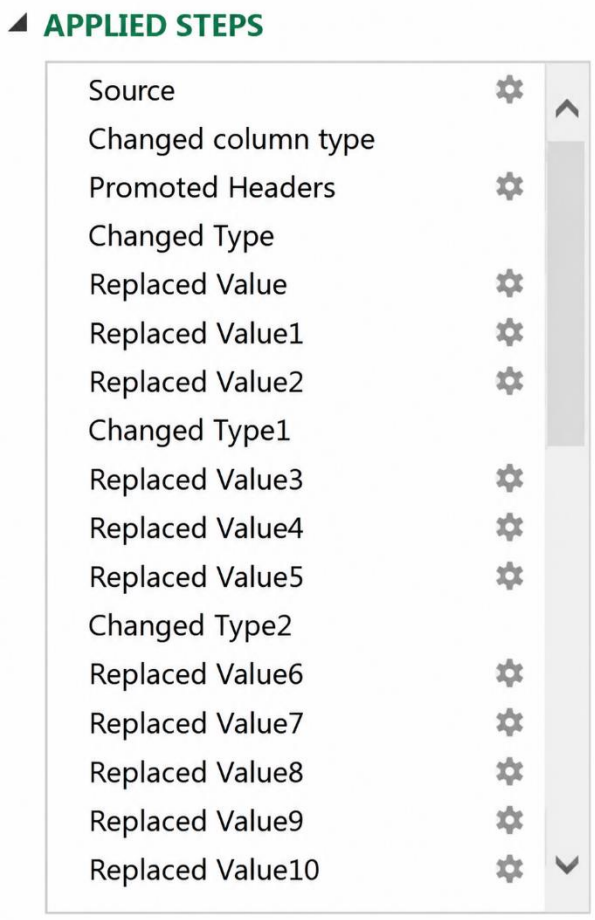
3.4 Text Cleaning

Text columns were cleaned using the **Trim** operation to remove unnecessary spaces and make the entries consistent.

3.5 Data Types

The following data types were assigned:

Column	Data Type
Transaction ID	Text
Item	Text
Quantity	Whole Number
Price Per Unit	Decimal Number
Total Spent	Decimal Number
Payment Method	Text
Location	Text
Transaction Date	Date



4. Result

After applying the cleaning operations in Power Query, the dataset became more consistent and structured. Missing and invalid entries were handled appropriately, duplicate records were removed, text values were standardized, and the correct data types were assigned.
The cleaned dataset is now ready for further analysis and visualization.

Sandwich	2	4	8	Unknown	Unknown
Sandwich	5	4	20	Cash	In-store
Smoothie	3	4	*null*	Unknown	Unknown
Cookie	5	1	5	Credit Card	In-store
Juice	4	3	12	Credit Card	Takeaway
Smoothie	5	4	20	Credit Card	Unknown
Cake	4	3	12	Digital Wallet	Takeaway
unknown	5	2	10	Digital Wallet	Unknown
unknown	2	1	*null*	Credit Card	Unknown
Smoothie	4	4	16	Unknown	In-store
unknown	5	1	5	Cash	Unknown
Smoothie	2	4	8	Digital Wallet	Takeaway
Sandwich	1	4	4	Credit Card	Unknown
unknown	4	3	12	Unknown	In-store
Salad	2	5	10	Credit Card	In-store

5. Conclusion

Power Query was successfully used to clean and prepare the customer sales dataset. The cleaning process improved the quality and consistency of the data and made it suitable for reliable analysis.